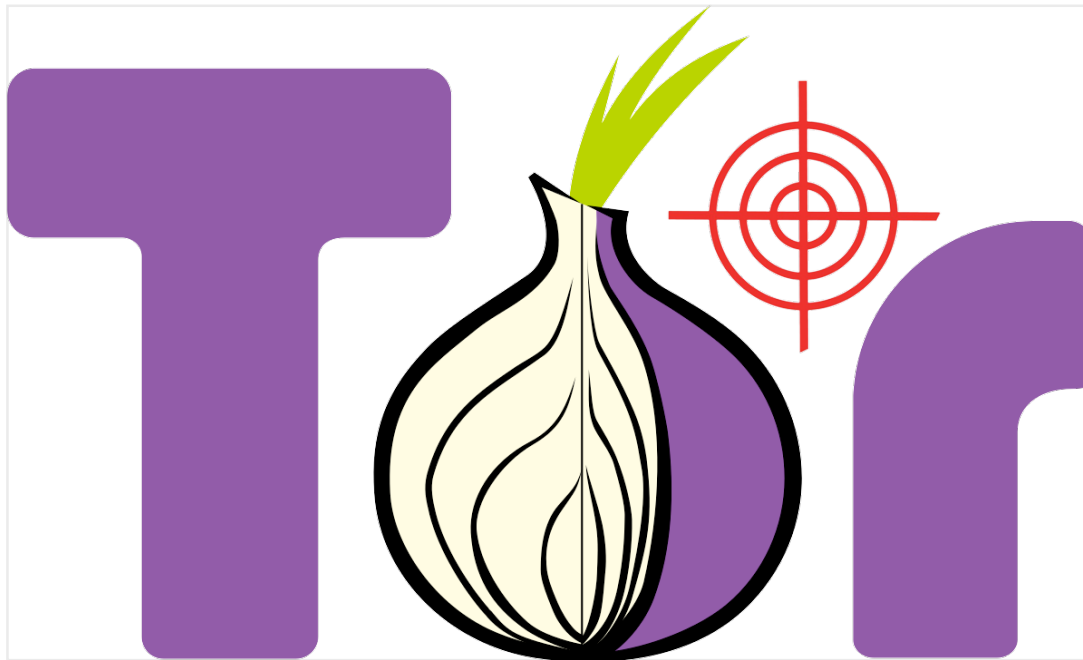


James Locklear Threat Hunt Final Project



Threat Hunt Report: Unauthorized TOR Usage

Platforms and Languages Leveraged

- Windows 11 Virtual Machines (Microsoft Azure)
- EDR Platform: Microsoft Defender for Endpoint
- Kusto Query Language (KQL)
- Tor Browser
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Scenario

Management suspects that some employees may be using TOR browsers to bypass network security controls because recent network logs show unusual encrypted traffic patterns and connections to known TOR entry nodes. Additionally, there have been anonymous reports of employees discussing ways to access restricted sites during work hours. The goal is to detect any TOR usage and analyze related security incidents to mitigate potential risks. If any use of TOR is found, notify management.

High-Level TOR-Related IoC Discovery Plan

- **Check DeviceFileEvents** for any tor(.exe) or firefox(.exe) file events.
- **Check DeviceProcessEvents** for any signs of installation or usage.
- **Check DeviceNetworkEvents** for any signs of outgoing connections over known TOR ports.

Steps Taken

1. Searched the DeviceFileEvents Table

Searched for any file that had the string "tor" in it and discovered what looks like the user "employee" downloaded a TOR installer, did something that resulted in many TOR-related files being copied to the desktop, and the creation of a file called tor-shopping-list.txt on the desktop at 2026-06-29T22:58:16.7259964Z. These events began at 2026-06-29T22:47:59.6065231Z.

Query used to locate the events, the time of the events, the account that initiated the breach, the device used, and the files created:

DeviceFileEvents

| where DeviceName == "threat-hunt-pro"
| where InitiatingProcessAccountName == "virtualuser"
| where FileName contains "tor"
| order by Timestamp desc
| project Timestamp, DeviceName, ActionType, FileName, FolderPath, SHA256, Account = InitiatingProcessAccountName

Query							
Run query							
Results							
Query history							
Getting started							
Export							
Show empty columns							
20 items							
00:02:804							
Low							
Last 7 days							
Add filter							
Timestamp	DeviceName	ActionType	FileName	FolderPath	SHA256	Account	
Jun 29, 2026 11:17...	threat-hunt-pro	FileCreated	tor.txt	C:\Users\virtualuser\Des...	5dc29ea302cc75db6...	virtualuser	
Jun 29, 2026 11:17...	threat-hunt-pro	FileCreated	Torbutton.txt	C:\Users\virtualuser\Des...	73b2a217734a2b37e...	virtualuser	
Jun 29, 2026 11:17...	threat-hunt-pro	FileCreated	Tor-Launcher.txt	C:\Users\virtualuser\Des...	d4fd98d01e1714e6b...	virtualuser	
Jun 29, 2026 11:13...	threat-hunt-pro	FileCreated	tor-shopping-list.lnk	C:\Users\virtualuser\AppData...	8b84de5867178c12e...	virtualuser	
Jun 29, 2026 11:13...	threat-hunt-pro	FileCreated	tor-shopping-list.txt	C:\Users\virtualuser\Des...	8d248bfc66cb368d9...	virtualuser	
Jun 29, 2026 10:59...	threat-hunt-pro	FileCreated	storage-sync-v2.sqlite	C:\Users\virtualuser\Des...		virtualuser	
Jun 29, 2026 10:59...	threat-hunt-pro	FileCreated	storage.sqlite	C:\Users\virtualuser\Des...		virtualuser	
Jun 29, 2026 10:58...	threat-hunt-pro	FileCreated	Tor Browser.lnk	C:\Users\virtualuser\Des...	8cbc8f15e977be2273...	virtualuser	
Jun 29, 2026 10:58...	threat-hunt-pro	FileCreated	tor.exe	C:\Users\virtualuser\Des...	ec7708e0b43e0e00b...	virtualuser	
Jun 29, 2026 10:58...	threat-hunt-pro	FileCreated	tor.txt	C:\Users\virtualuser\Des...	5dc29ea302cc75db6...	virtualuser	
Jun 29, 2026 10:58...	threat-hunt-pro	FileCreated	Torbutton.txt	C:\Users\virtualuser\Des...	73b2a217734a2b37e...	virtualuser	
Jun 29, 2026 10:58...	threat-hunt-pro	FileCreated	Tor-Launcher.txt	C:\Users\virtualuser\Des...	d4fd98d01e1714e6b...	virtualuser	
Jun 29, 2026 10:58...	threat-hunt-pro	FileDeleted	tor-browser-windows-x8...	C:\Users\virtualuser\Do...		virtualuser	
Jun 29, 2026 10:51...	threat-hunt-pro	FileRenamed	tor-browser-windows-x8...	C:\Users\virtualuser\Do...	13d3e0a229443b643...	virtualuser	
Jun 29, 2026 10:51...	threat-hunt-pro	FileModified	edge_tracking_page_vali...	C:\Windows\SystemTem...		virtualuser	
Jun 29, 2026 10:51...	threat-hunt-pro	FileModified	edge_confirmation_pag...	C:\Windows\SystemTem...		virtualuser	
Jun 29, 2026 10:51...	threat-hunt-pro	FileModified	edge_checkout_page_va...	C:\Windows\SystemTem...		virtualuser	

2. Searched the DeviceProcessEvents Table

Searched for any ProcessCommandLine that contained the string "tor-browser-

windows-x86_64-portable-15.0.17.exe". Based on the logs returned, at 2026-06-29T22:51:20.4484567Z, an employee on the "threat-hunt-pro" VM device ran the file tor-browser-windows-x86_64-portable-15.0.17.exe from their Downloads folder, using a command that triggered a silent installation.

Query used:
DeviceProcessEvents

| where DeviceName == "threat-hunt-lab"
| where ProcessCommandLine contains "tor-browser-windows-x86_64-portable-15.0.17.exe"
| project Timestamp, DeviceName, AccountName, ActionType, FileName, FolderPath, SHA256, ProcessCommandLine

The screenshot shows a SIEM query results interface. At the top, there's a 'Query' section with a 'Run query' button and options to 'Create summary rule' and 'Create detection rule'. Below this is a 'Results' section with a search bar and a table of results. The table has columns for Timestamp, DeviceName, AccountName, ActionType, FileName, FolderPath, SHA256, and ProcessCommandLine. Two items are listed, both for 'ProcessCreated' events on 'threat-hunt-pro' device by 'virtualuser'.

Timestamp	DeviceName	AccountName	ActionType	FileName	FolderPath	SHA256	ProcessCommandLine
Jun 29, 2026 10:58...	threat-hunt-pro	virtualuser	ProcessCreated	tor-browser-windows-x8...	C:\Users\virtualuser\Do...	13d3e0a229443b643...	"tor-browser-window
Jun 29, 2026 11:17...	threat-hunt-pro	virtualuser	ProcessCreated	tor-browser-windows-x8...	C:\Users\virtualuser\Do...	13d3e0a229443b643...	"tor-browser-window

3. Searched the DeviceProcessEvents Table for TOR Browser Execution

Searched for any indication that user "virtualuser" actually opened the TOR browser. There was evidence that they did open it at 2026-06-29T10:59:28.6357935Z. There were several other instances of firefox.exe (TOR) as well as tor.exe spawned afterwards.

Query used to locate events:
DeviceProcessEvents

| where DeviceName == "threat-hunt-pro"
| where FileName has_any ("tor.exe", "firefox.exe", "tor-browser.exe")
| project Timestamp, DeviceName, AccountName, ActionType, FileName, FolderPath, SHA256, ProcessCommandLine
| order by Timestamp desc

Query

Run query

Results

Query history

Getting started

Export

23 items

Search

00:01:281

Low

Filters:

Add filter

☐	Timestamp	☐	DeviceName	☐	AccountName	☐	ActionType	☐	FileName	☐	FolderPath	☐	SHA256	☐	ProcessCommandL...
☐	>	Jun 29, 2026 11:03...	threat-hunt-pro	virtualuser	ProcessCreated	firefox.exe	C:\Users\virtualuser\Des...	241366fae863e43e7a...	"firefox.exe" -cont						
☐	>	Jun 29, 2026 11:03...	threat-hunt-pro	virtualuser	ProcessCreated	firefox.exe	C:\Users\virtualuser\Des...	241366fae863e43e7a...	"firefox.exe" -cont						
☐	>	Jun 29, 2026 11:01...	threat-hunt-pro	virtualuser	ProcessCreated	firefox.exe	C:\Users\virtualuser\Des...	241366fae863e43e7a...	"firefox.exe" -cont						
☐	>	Jun 29, 2026 11:01...	threat-hunt-pro	virtualuser	ProcessCreated	firefox.exe	C:\Users\virtualuser\Des...	241366fae863e43e7a...	"firefox.exe" -cont						
☐	>	Jun 29, 2026 11:01...	threat-hunt-pro	virtualuser	ProcessCreated	firefox.exe	C:\Users\virtualuser\Des...	241366fae863e43e7a...	"firefox.exe" -cont						
☐	>	Jun 29, 2026 11:01...	threat-hunt-pro	virtualuser	ProcessCreated	firefox.exe	C:\Users\virtualuser\Des...	241366fae863e43e7a...	"firefox.exe" -cont						
☐	>	Jun 29, 2026 11:00...	threat-hunt-pro	virtualuser	ProcessCreated	firefox.exe	C:\Users\virtualuser\Des...	241366fae863e43e7a...	"firefox.exe" -cont						
☐	>	Jun 29, 2026 11:00...	threat-hunt-pro	virtualuser	ProcessCreated	firefox.exe	C:\Users\virtualuser\Des...	241366fae863e43e7a...	"firefox.exe" -cont						
☐	>	Jun 29, 2026 11:00...	threat-hunt-pro	virtualuser	ProcessCreated	firefox.exe	C:\Users\virtualuser\Des...	241366fae863e43e7a...	"firefox.exe" -cont						
☐	>	Jun 29, 2026 10:59...	threat-hunt-pro	virtualuser	ProcessCreated	tor.exe	C:\Users\virtualuser\Des...	ec7708e0b43e0e00b...	"tor.exe" -f "C:\Use						
☐	>	Jun 29, 2026 10:59...	threat-hunt-pro	virtualuser	ProcessCreated	firefox.exe	C:\Users\virtualuser\Des...	241366fae863e43e7a...	"firefox.exe" -cont						
☐	>	Jun 29, 2026 10:59...	threat-hunt-pro	virtualuser	ProcessCreated	firefox.exe	C:\Users\virtualuser\Des...	241366fae863e43e7a...	"firefox.exe" -cont						
☐	>	Jun 29, 2026 10:59...	threat-hunt-pro	virtualuser	ProcessCreated	firefox.exe	C:\Users\virtualuser\Des...	241366fae863e43e7a...	"firefox.exe" -cont						
☐	>	Jun 29, 2026 10:59...	threat-hunt-pro	virtualuser	ProcessCreated	firefox.exe	C:\Users\virtualuser\Des...	241366fae863e43e7a...	"firefox.exe" -cont						
☐	>	Jun 29, 2026 10:59...	threat-hunt-pro	virtualuser	ProcessCreated	firefox.exe	C:\Users\virtualuser\Des...	241366fae863e43e7a...	"firefox.exe" -cont						
☐	>	Jun 29, 2026 10:59...	threat-hunt-pro	virtualuser	ProcessCreated	firefox.exe	C:\Users\virtualuser\Des...	241366fae863e43e7a...	"firefox.exe" -cont						

4. Searched the DeviceNetworkEvents Table for TOR Network Connections

Searched for any indication the TOR browser was used to establish a connection using any of the known TOR ports. At 2026-06-29T22:59:49.1246358Z, an employee on the "threat-hunt-pro" VM device successfully established a connection to the remote IP address 212.132.97.196 on remote port 443. The connection was initiated by the process tor.exe, located in the folder c:\users\employee\desktop\tor browser\browser\torbrowser\tor\tor.exe. There were a couple of other connections to sites over remote ports 9001 and 9150 with IP addresses of 94.130.133.51 and 127.0.0.1

Query used to locate events:

DeviceNetworkEvents

| where DeviceName == "threat-hunt-pro"

| where InitiatingProcessAccountName != "system"

| where InitiatingProcessFileName in ("tor.exe", "firefox.exe")

| where RemotePort in ("9001", "9030", "9040", "9050", "9051", "9150", "80", "443")

| project Timestamp, DeviceName, InitiatingProcessAccountName, ActionType, RemoteIP, RemotePort, RemoteUrl, InitiatingProcessFileName, InitiatingProcessFolderPath

| order by Timestamp desc

Results

Query history

Getting started

Export

Show empty columns

5 items

Search

00:01.915

Low

Filters:

Add filter

Timestamp	DeviceName	InitiatingProcessAc...	ActionType	RemoteIP	RemotePort	RemoteUrl	InitiatingProcessFil...
> Jun 29, 2026 10:59... threat-hunt-pro virtualuser ConnectionSuccess 127.0.0.1 9150 firefox.exe							
> Jun 29, 2026 10:59... threat-hunt-pro virtualuser ConnectionSuccess 94.130.133.51 9001 https://www.odmouu... tor.exe							
> Jun 29, 2026 10:59... threat-hunt-pro virtualuser ConnectionSuccess 94.130.133.51 9001 tor.exe							
> Jun 29, 2026 10:59... threat-hunt-pro virtualuser ConnectionSuccess 212.132.97.196 443 https://www.5mxnlyq... tor.exe							
> Jun 29, 2026 10:59... threat-hunt-pro virtualuser ConnectionSuccess 212.132.97.196 443 tor.exe							

Chronological Event Timeline

1. File Download - TOR Installer

- **Timestamp:** 2026-06-29T22:47:59.6065231Z
- **Event:** The user "virtual user" downloaded a file named tor-browser-windows-x86_64-portable-15.0.17.exe to the Downloads folder.
- **Action:** File download detected.
- **File Path:** C:\Users\employee\Downloads\ttor-browser-windows-x86_64-portable-15.0.17.exe
-

2. Process Execution - TOR Browser Installation

- **Timestamp:** 2026-06-29T22:51:20.4484567Z
- **Event:** The user "virtual user" executed the file tor-browser-windows-x86_64-portable-15.0.17.exe in silent mode, initiating a background installation of the TOR Browser.
- **Action:** Process creation detected.
- **Command:** tor-browser-windows-x86_64-portable-15.0.17.exe /S
- **File Path:** C:\Users\employee\Downloads\ttor-browser-windows-x86_64-portable-15.0.17.exe
-

3. Process Execution - TOR Browser Launch

- **Timestamp:** 2026-06-29T10:59:28.6357935Z
- **Event:** User "virtualuser" opened the TOR browser. Subsequent processes associated with TOR browser, such as firefox.exe and tor.exe, were also created, indicating that the browser launched successfully.
- **Action:** Process creation of TOR browser-related executables detected.
- **File Path:** C:\Users\employee\Desktop\Tor Browser\Browser\TorBrowser\Tor\tor.exe
-

4. Network Connection - TOR Network

- **Timestamp:** 2026-06-29T22:59:49.1246358Z
- **Event:** A network connection to IP 212.132.97.196 on port 443 by user

"virtualuser" was established using tor.exe, confirming TOR browser network activity.

- **Action:** Connection success.
- **Process:** tor.exe
- **File Path:** c:\users\employee\desktop\tor browser\browser\torbrowser\tor\tor.exe

-

5. Additional Network Connections - TOR Browser Activity

- **Timestamps:**
 - 2026-06-29T22:59:52Z - Connected to 94.130.133.51 on remote port 9001.
 - 2026-06-29T22:59:59Z - Local connection to 127.0.0.1 on remote port 9150.
- **Event:** Additional TOR network connections were established, indicating ongoing activity by user "virtualuser" through the TOR browser.
- **Action:** Multiple successful connections detected.

-

6. File Creation - TOR Shopping List

- **Timestamp:** 2026-06-29T22:58:16.7259964Z
- **Event:** The user "virtualuser" created a file named tor-shopping-list.txt on the desktop, potentially indicating a list or notes related to their TOR browser activities.
- **Action:** File creation detected.
- **File Path:** C:\Users\employee\Desktop\tor-shopping-list.txt

Summary

The user "virtuluser" on the "threat-hunt-pro" VM device initiated and completed the installation of the TOR browser. They proceeded to launch the browser, establish connections within the TOR network, and created various files related to TOR on their desktop, including a file named "tor-shopping-list.txt." This sequence of activities indicates that the user actively installed, configured, and used the TOR browser, likely for anonymous browsing purposes, with possible documentation in the form of the "shopping list" file.

Response Taken

TOR usage was confirmed on the endpoint threat-hunt-pro by the user "virtualuser". The VM device threat-hunt-pro device was isolated, and the user's direct manager was notified.